ESP32-S3 SERVICE / RECOVERY CANDIDATE — NO MOTION OR ACCESSORY OUTPUT AUTHORITY

Self-powered USB2 device; USB VBUS never powers controller. USB connector is an electrical boundary pending exact pad mapping.
U2 is Schmitt AND: module EN = sequencer ESP_CHIP_PU AND manual-reset release. Never ground sequencer output.
Host RX is GPIO4 via powered-off buffer; default firmware leaves it disabled. UART0 adapter VREF powers only its outgoing buffer.

U1
ESP32-S3-WROOM-1-N8
LOGIC_GND 1 GND
ESP_3V3 2 3V3
ESP_MODULE_EN 3 EN
ESP_HOST_RX 4 IO4
ESP_RESERVED_GPIO5 5 IO5
ESP_RESERVED_GPIO6 6 IO6
ESP_RESERVED_GPIO7 7 IO7
8 IO15
9 IO16
10 IO17
11 IO18
ESP_RESERVED_GPIO8 12 IO8
ESP_USB_DM 13 IO19_USB_DM
ESP_USB_DP 14 IO20_USB_DP
ESP_STRAP3 15 IO3
ESP_STRAP46 16 IO46
ESP_RESERVED_GPIO9 17 IO9
ESP_RESERVED_GPIO10 18 IO10
ESP_RESERVED_GPIO11 19 IO11
ESP_RESERVED_GPIO12 20 IO12
21 IO13
22 IO14
23 IO21
24 IO47
25 IO48
ESP_STRAP45 26 IO45
ESP_BOOT_N 27 IO0
28 IO35
29 IO36
30 IO37
31 IO38
32 IO39
33 IO40
34 IO41
35 IO42
ESP_UART0_RX 36 RXD0_IO44
ESP_UART0_TX 37 TXD0_IO43
38 IO2
39 IO1
LOGIC_GND 40 GND
LOGIC_GND 41 GND_PAD

U2
SN74LVC1G97DBVR
LOGIC_GND 1 IN1
LOGIC_GND 2 GND
ESP_MANUAL_RESET_N 3 IN0
4 Y
ESP_3V3 5 VCC
ESP_CHIP_PU 6 IN2
ESP_MODULE_EN

U3
SN74LVC2G17DBVR
HOST_HEALTH_RX_FILTERED 1 1A
LOGIC_GND 2 GND
SERVICE_UART_RX_FILTERED 3 2A
4 2Y
ESP_3V3 5 VCC
ESP_HOST_RX 6 1Y
ESP_UART0_RX

U4
SN74LVC1G17DBVR
1 NC
ESP_UART0_TX 2 A
LOGIC_GND 3 GND
4 Y
ESP_3V3 5 VCC
SERVICE_VREF_3V3 6 NC
SERVICE_UART_TX_BUFFERED

U5
TS3USB221ARSER
ESP_USB_DP_SWITCH 1 1D_PLUS
ESP_USB_DM_SWITCH 2 1D_MINUS
3 2D_PLUS
4 2D_MINUS
LOGIC_GND 5 GND
6 OE_N
USB_DM_CONNECTOR 7 D_MINUS
USB_DP_CONNECTOR 8 D_PLUS
LOGIC_GND 9 S
ESP_3V3 10 VCC
ESP_USB_OE_N

U6
SN74LVC1G97DBVR
ESP_3V3 1 IN1
LOGIC_GND 2 GND
LOGIC_GND 3 IN0
4 Y
ESP_3V3 5 VCC
USB_VBUS_SENSE 6 IN2
ESP_USB_OE_N

D1
TPD2EUSB30DRTR
USB_DP_CONNECTOR 1 D_PLUS
USB_DM_CONNECTOR 2 D_MINUS
3 GND
LOGIC_GND

D2
TPD1E10B06DPYR
USB_VBUS 1 IO
2 IO
LOGIC_GND

D3
TPD1E10B06DPYR
USB_CC1 1 IO
2 IO
LOGIC_GND

D4
TPD1E10B06DPYR
USB_CC2 1 IO
2 IO
LOGIC_GND

R1 10kΩ ESP_3V3 — ESP_BOOT_N
R2 10kΩ ESP_STRAP45 — LOGIC_GND
R3 10kΩ ESP_STRAP46 — LOGIC_GND
R4 10kΩ ESP_STRAP3 — LOGIC_GND
R5 10kΩ ESP_3V3 — ESP_MANUAL_RESET_N
R6 47kΩ ESP_MODULE_EN — LOGIC_GND
R7 1kΩ HOST_HEALTH_TX_3V3 — HOST_HEALTH_RX_FILTERED
R8 100kΩ HOST_HEALTH_RX_FILTERED — LOGIC_GND
R9 1kΩ SERVICE_ADAPTER_TX_3V3 — SERVICE_UART_RX_FILTERED
R10 100kΩ SERVICE_UART_RX_FILTERED — LOGIC_GND
R11 100kΩ SERVICE_UART_TX_BUFFERED — SERVICE_ADAPTER_RX_3V3
R12 22Ω USB_DP — ESP_USB_DP_SWITCH
R13 22Ω USB_DM — ESP_USB_DM_SWITCH
R14 10kΩ ESP_3V3 — ESP_USB_OE_N
R15 10kΩ USB_VBUS — USB_VBUS_SENSE
R16 100kΩ USB_VBUS_SENSE — LOGIC_GND
R17 5.1kΩ USB_CC1 — LOGIC_GND
R18 5.1kΩ USB_CC2 — LOGIC_GND
R19 0Ω USB_SHIELD — LOGIC_GND

C1 22uF16V 20% X5R 0402 — LOGIC_GND
C2 100nF16V X7R 0402 — LOGIC_GND
C3 100nF16V X7R 0402 — LOGIC_GND
C4 100nF16V X7R 0402 — LOGIC_GND
C5 100nF16V X7R 0402 — LOGIC_GND
C6 100nF16V X7R 0402 — LOGIC_GND
C7 100nF16V X7R 0402 — LOGIC_GND
C8 100nF16V X7R 0402 — LOGIC_GND

#FLG0 PWR_FLAG — ESP_3V3
#FLG1 PWR_FLAG — LOGIC_GND
#FLG2 PWR_FLAG — SERVICE_VREF_3V3

J1
USB-C ELECTRICAL BOUNDARY
LOGIC_GND A1
USB_VBUS A4
USB_CC1 A5
USB_DP_CONNECTOR A6
USB_DM_CONNECTOR A7
A8
USB_VBUS A9
LOGIC_GND A12
LOGIC_GND B1
B4 USB_VBUS
B5 USB_CC2
B6 USB_DP_CONNECTOR
B7 USB_DM_CONNECTOR
B8
B9 USB_VBUS
B12 LOGIC_GND
S1 USB_SHIELD

JRAIL1
RAIL BOUNDARY
ESP_3V3 1
LOGIC_GND 2
3 ESP_CHIP_PU

JHOST1
HOST BOUNDARY
HOST_HEALTH_TX_3V3 1
2 LOGIC_GND

JSERIAL1
SERIAL BOUNDARY
SERVICE_VREF_3V3 1
SERVICE_ADAPTER_TX_3V3 2
3 SERVICE_ADAPTER_RX_3V3
4 LOGIC_GND

JRESET1
RESET BOUNDARY
ESP_MANUAL_RESET_N 1
2 LOGIC_GND

JBOOT1
BOOT BOUNDARY
ESP_BOOT_N 1
2 LOGIC_GND

JRES1
RES BOUNDARY
ESP_RESERVED_GPIO5 1
ESP_RESERVED_GPIO6 2
ESP_RESERVED_GPIO7 3
ESP_RESERVED_GPIO8 4
ESP_RESERVED_GPIO9 5
6 ESP_RESERVED_GPIO10
7 ESP_RESERVED_GPIO11
8 ESP_RESERVED_GPIO12
9 LOGIC_GND

All footprints unassigned: no board layout or manufacture claim. USB switch D pins face connector: SD pins face ESP: port2NC andSground.
Two independentR.14 CC resistors. SBU pins NC. NoPD/High-voltage mode. Local USB shield bond toLOGIC_GND is provisional: assess enclosure/EMC.
JRESET/JBOOT: normally-open dry contacts only. HoldBOOT low, holdRESET>=20ms, releaseRESET, keepBOOTlow>=3ms thenrelease.
JRES is reserved internal3.3V logic only: current firmware provides no accessory support. NoSV/24V/unpowered external drive.
Module antenna outsidecarrieredge preferred; enclosureclearance>=15mm all directions. Shared2.4GHzWi-Fi/BLE, noSGHzorClassicBT.

Sheet: /
File: esp_service.kicad_sch
Title: ESP service and recovery
Size: A1 | Date: 2026-09-28 | Rev: r0.1
KiCad E.D.A. 10.0.4 | Id: 1/1